

SUDARSHAN UAV — RELEASE GATE CHECKLIST

Pre-Release Validation | Complete ALL gates before deployment

This checklist must be completed in full before any new firmware or software version is approved for flight operations. Each item must be verified by a qualified engineer. Incomplete items constitute a release blocker.

Version / Tag:	_____	Release Date:	_____
Branch:	_____	Build SHA:	_____

GATE 1 — Code Quality

■ 1.	pytest suite passes (0 failures)	Run: <code>pytest tests/ -v</code> — all tests must show PASSED	PASS / FAIL	_____
■ 2.	Syntax check (flake8 / pylint) clean	Run: <code>flake8 GCS/ FC/</code> — zero E/W errors	PASS / FAIL	_____
■ 3.	No new TODO without linked issue	<code>grep -r 'TODO'</code> — each must have <code>#issue-number</code>	PASS / FAIL	_____
■ 4.	CHANGELOG.md updated with this release	Version entry with date, changes, and breaking notes	PASS / FAIL	_____
■ 5.	Version string bumped in firmware + GCS	Verify <code>VERSION</code> constant in Mega FC and GCS config match tag	PASS / FAIL	_____
■ 6.	PROTOCOL.md matches all implemented commands	Manually diff command list in code vs <code>PROTOCOL.md</code> — no gaps	PASS / FAIL	_____

GATE 2 — Firmware Bench Validation

■ 1.	All 3 boards flashed with release firmware	Mega FC, Uno motor driver, ESP32 gateway	Record flash SHA	_____
■ 2.	Serial monitor shows READY on Mega + Open	serial at 115200; confirm within 5 s of power-on	PASS / FAIL	_____
■ 3.	GCS connects and session established	TCP connect to 192.168.4.1:5760; green indicator	PASS / FAIL	_____
■ 4.	Telemetry flows at 10 Hz	GCS log shows seq incrementing; no freezes over 30 s	PASS / FAIL	_____
■ 5.	MOTOR_TEST — all 4 motors respond in <code>CH0 FL</code>	→ <code>CH1 FR</code> → <code>CH2 RL</code> → <code>CH3 RR</code> ; each spins 0.5 s at 15%	PASS / FAIL	_____
■ 6.	ARM → HOVER nominal (props on, restrained)	Issue <code>ARM</code> , <code>HOVER</code> command; confirm altitude hold; check <u>oscillation</u>	PASS / FAIL	_____
■ 7.	LAND → auto-DISARM confirmed	Issue <code>LAND</code> ; verify state transitions <code>HOVER</code> → <code>LAND</code> → <code>DISARMED</code>	PASS / FAIL	_____
■ 8.	DMS test: disconnect GCS for 30 s	Disconnect TCP; drone must auto-LAND within 35 s; reconnect and verify	PASS / FAIL	_____
■ 9.	KILL command confirmed	During HOVER (restrained), issue <code>KILL</code> ; confirm immediate <u>motor cutoff</u>	PASS / FAIL	_____
■ 10.	Battery voltage matches multimeter	Compare <code>bat_mv</code> to handheld multimeter reading; error < 100 mV	PASS / FAIL	_____

GATE 3 — Security Review

■ 1.	<code>credentials.h</code> and <code>credentials.py</code> NOT in <code>git</code>	status: committed files — no commit history <code>grep credentials</code> must be empty	PASS / FAIL	_____
■ 2.	No hardcoded IP addresses or credentials	<code>grep -r '192.168\ password\ api_key' src/</code> — zero results	PASS / FAIL	_____

■ 3.	AES key not logged to serial or telemetry	Code review: AES key variable never passed to Serial.print or telemetry_struct	PASS-FAIL
■ 4.	Rate limiting functional	Send 20 rapid commands in 1 s; verify GCS receives RATE_LIMIT response	PASS-FAIL

GATE 4 — Documentation

■ 1.	PROTOCOL.md is current with all FC commands	Compare PROTOCOL.md command table to FC command parser — no missing entries	PASS-FAIL
■ 2.	WIRING.md is current with actual hardware	Compare WIRING.md to physical build; all pin assignments accurate	PASS-FAIL
■ 3.	Project report PDF regenerated for this release	Re-gen_report.py; verify PDF timestamp matches release date	PASS-FAIL
■ 4.	CHANGELOG.md has entry for this version	Release section present with: version, date, changes, known issues	PASS-FAIL

Release Sign-Off

Role	Printed Name	Signature	Date
Bench Test & Validation		_____	_____
Code Review		_____	_____
Release Approved By	<i>Sudarshan Project Director</i>	_____	_____

All four gates must be signed off before this version is cleared for flight operations. Any FAIL result in Gates 1–4 is a release blocker and must be resolved and re-verified.